II-Mid Semester Examination **Subject – Digital Electronics (CSA102)**

Date: 22.12.2023 Max. Marks: 10 Duration: 1 hour **Set - B**

		Marks	CO	BL
Q.1	Convert the following expression to sum of product form and draw logic diagram. $(X+Z)(X\bar{Y}+XZ)(\bar{X}Z+\bar{Y})$ **OR**	03	CO2	BL3
	Simplify the following Boolean expression and realize the minimised circuit diagram. $Z=\overline{A\bar{B}.(A+C)}+\bar{A}B.\overline{A+\bar{B}+\bar{C}}$	03	CO2	BL3
Q.2	Draw a Karnaugh map for the following function of four variables. $F(A,B,C,D)=\sum m(3,5,6,7,8,11,12,13,14,15)$ **OR**	04	CO3	BL4
	Minimize the term $\sum m(0,1,2,4,5,10,12,13,14,15)+\sum d(6,8)$ using Quine McCluskey method.	04	CO3	BL4
Q.3	What is combinational circuit? Design a half-subtract combinational circuit. **OR**	03	CO4	BL6
	What is Flip-Flop? Why we need it? How can you make D flip flop from RS flip flop?	03	CO4	BL6

Mid Sem Exam –II Dec. 2023

Class: B.Tech I Sem Date: 28.12.23 Branch: IT Room: 002 & 005

Time: 12.30 to 1.30 PM Sub: Applied Chemistry Set: A MM 10

Note: Attempt any 2 questions:

Q 1.(i) Derive the Nernst equation for single electrode potential? [3]

(ii) Give the charging and discharging reactions of Pb-acid battery?[2]

Q.2 (i)Write the short note on Hot Dipping method with diagram? [3]

(ii) Define chemical corrosion with example? [2]

Q.3 (i) Write the classification of Batteries with example? [3]

(ii) Write the short note on Graphene and Na-ion battery? [2]

Q.4(i)Briefly explain the mechanism of Wet corrosion with reactions? [3](ii)Write the factors affecting the rate of corrosion? [2]

Q.	COs	BL
1	CO2	2/3
2	CO3	1/2
3	CO2	2/4
4	CO3	1/2